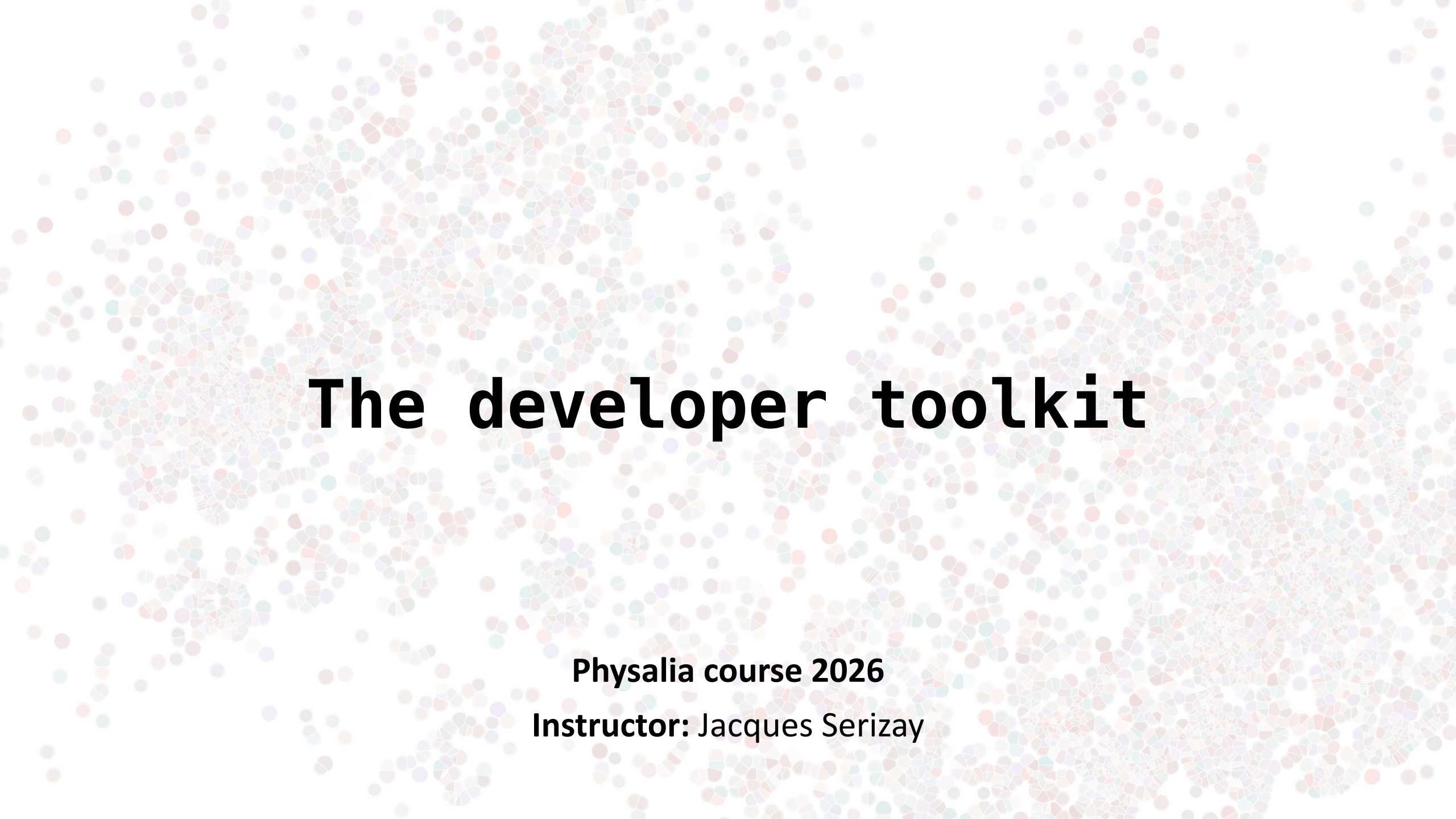




# **The developer toolkit**

**Physalia course 2026**

**Instructor: Jacques Serizay**

To develop an R/Bioc package, developers must be careful of the weather...

<https://contributions.bioconductor.org/use-devel.html#use-devel>

Package authors should develop against the version of R that will be available to users when the Bioconductor devel branch becomes the Bioconductor release branch.

R has a '.y' release in x.y.z every year (typically mid-April), but Bioconductor has a .y release (where current devel becomes release) every 6 months (mid-April and mid-October).

This means that:

- from mid-October through mid-April, Bioconductor developers should be developing against R-devel
- from mid-April to mid-October, developers should use R-release for Bioconductor development.

To develop an R/Bioc package, developers must be careful of the

<https://contributions.bioconductor.org/use-de>

Package authors should  
when the R/Bioc

**BEWARE  
R/BIOC VERSION**

as a .y  
pml and mid-

- In April, Bioconductor developers should be developing  
and-October, developers should use R-release  
auctor development.

# The (Bioc.) developer toolkit

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<https://r-pkgs.org/>

<https://contributions.bioconductor.org/index.html>

devtools

usethis

remotes

testthat

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RStudio

➤ And many more...: <https://indrajeetpatil.github.io/awesome-r-pkgtools/>

devtools makes package development easier by providing R functions that simplify and expedite common tasks.

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- For interactive use, users should attach devtools and think of it as the provider of your favorite functions for package development.

```
library(devtools)  
#Loading required package: usethis  
load\_all()
```



devtools makes package development easier by providing R functions that simplify and expedite common tasks.

- For interactive use, users should attach devtools and think of it as the provider of your favorite functions for package development.

```
library(devtools)  
#Loading required package: usethis  
load_all()
```

- For programmatic use, such as inside another package, developers should **NOT** depend on devtools, but should instead access functions via the package that is their primary home.

```
pkgload::load_all()
```

devtools makes package development easier by providing R functions that simplify and expedite common tasks.

devtools' NAMESPACE:

```
importFrom(cli,cat_bullet)
importFrom(cli,cat_rule)
importFrom(ellipsis,check_dots_used)
importFrom(lifecycle,deprecated)
importFrom(memoise,memoise)
importFrom(miniUI,miniPage)
importFrom(pkgbuild,clean_dll)
importFrom(pkgbuild,find_rtools)
importFrom(pkgbuild,has_devel)
importFrom(pkgbuild,with_debug)
importFrom(pkgload,check_dep_version)
importFrom(pkgload,parse_deps)
importFrom(pkgload,unload)
importFrom(profvis,profvis)
importFrom(remotes,dev_package_deps)
importFrom(remotes,github_pull)
importFrom(remotes,github_release)
importFrom(remotes,install_bioc)
importFrom(remotes,install_bitbucket)
importFrom(remotes,install_cran)
importFrom(remotes,install_dev)
importFrom(remotes,install_git)
importFrom(remotes,install_github)
```

```
importFrom(remotes,install_gitlab)
importFrom(remotes,install_local)
importFrom(remotes,install_svn)
importFrom(remotes,install_url)
importFrom(remotes,install_version)
importFrom(remotes,update_packages)
importFrom(sessioninfo,package_info)
importFrom(sessioninfo,session_info)
importFrom(stats,update)
importFrom(urlchecker,url_check)
importFrom(usethis,ui_code)
importFrom(usethis,ui_done)
importFrom(usethis,ui_field)
importFrom(usethis,ui_path)
importFrom(usethis,ui_todo)
importFrom(usethis,ui_value)
importFrom(usethis,use_test)
importFrom(utils,available.packages)
importFrom(utils,contrib.url)
importFrom(utils,install.packages)
importFrom(utils,installed.packages)
importFrom(utils,modifyList)
importFrom(utils,packageDescription)
importFrom(utils,packageVersion)
importFrom(utils,remove.packages)
```

“One Ring to rule them all”  
– The Free People

# The Bioc. developer toolkit

<https://r-pkgs.org/>

<https://contributions.bioconductor.org/index.html>

devtools

**usethis** → Fastening early package development

remotes

testthat

BiocCheck

Biocthis

RStudio

➤ And many more...: <https://indrajeetpatil.github.io/awesome-r-pkgtools/>

# The Bioc. developer toolkit

<https://r-pkgs.org/>

<https://contributions.bioconductor.org/index.html>

devtools

usethis

**remotes** → Installing packages from anywhere

testthat

BiocCheck

Biocthis

RStudio

➤ And many more...: <https://indrajeetpatil.github.io/awesome-r-pkgtools/>

# Some useful functions

```
>remotes::install_*
```

```
remotes::install_gitlab  
remotes::install_github  
remotes::install_cran  
remotes::install_deps  
remotes::install_bitbucket
```

```
remotes::install_local  
remotes::install_dev  
remotes::install_url  
remotes::install_svn  
remotes::install_version  
remotes::install_bioc  
remotes::install_git
```

```
remotes::install_remote
```

```
>usethis::use_*
```

```
usethis::use_gpl_license  
usethis::use_git_ignore  
usethis::use_version  
usethis::use_roxygen_md  
usethis::use_github_action  
usethis::use_github  
usethis::use_data  
usethis::use_tibble  
usethis::use_vignette  
usethis::use_readme_md
```

```
usethis::use_description  
usethis::use_git_credentials  
usethis::use_test  
usethis::use_pipe  
usethis::use_agpl_license  
usethis::use_tidy_upkeep_issue  
usethis::use_github_links  
usethis::use_description_defaults  
usethis::use_bioc_badge  
usethis::use_pkgdown_github_pages  
usethis::use_tidy_github_actions  
usethis::use_coverage
```

```
usethis::use_package_doc  
usethis::use_github_actions  
usethis::use_git_config  
usethis::use_usethis  
usethis::use_pkgdown  
usethis::use_testthat
```

# The Bioc. developer toolkit

<https://r-pkgs.org/>

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devtools

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testthat

→ Adding unit testing in later package development

BiocCheck

Biocthis

RStudio

➤ And many more...: <https://indrajeetpatil.github.io/awesome-r-pkgtools/>

# The Bioc. developer toolkit

<https://r-pkgs.org/>

<https://contributions.bioconductor.org/index.html>

devtools

usethis

remotes

testthat

BiocCheck

**Biocthis** → Extra Bioconductor-specific features

RStudio

➤ And many more...: <https://indrajeetpatil.github.io/awesome-r-pkgtools/>

# Some useful functions

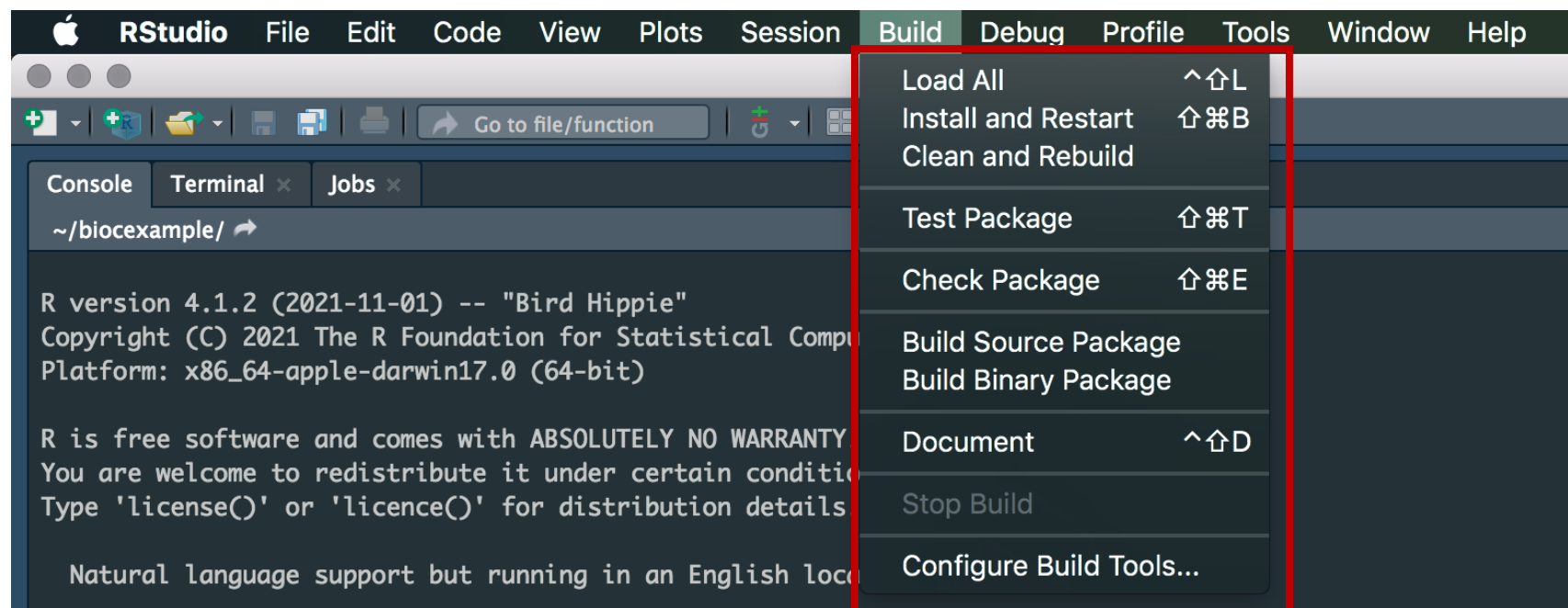
```
>testthat::expect_*  
  
testthat::expect_match  
testthat::expect_snapshot_warning  
testthat::expect_snapshot_error  
testthat::expect_equal  
testthat::expect_lt  
testthat::expect_cpp_tests_pass  
testthat::expect_condition  
testthat::expect_equivalent  
testthat::expect_failure  
testthat::expect_visible  
testthat::expect_s3_class  
testthat::expect_known_hash  
testthat::expect_length  
testthat::expect_less_than  
testthat::expect_equal_to_reference  
testthat::expect_warning  
  
testthat::expect_snapshot  
testthat::expect_snapshot_output  
testthat::expect_true  
testthat::expect_that  
testthat::expect_error  
testthat::expect_lte  
testthat::expect_reference  
testthat::expect_s4_class  
testthat::expect_setequal  
testthat::expect_invisible  
testthat::expect_null  
testthat::expect_message  
testthat::expect_snapshot_value  
testthat::expect_mapequal  
testthat::expect_silent  
testthat::expect_more_than  
testthat::expect_known_value  
testthat::expect_named  
  
testthat::expect_false  
testthat::expect_is  
testthat::expect_type  
testthat::expect_output_file  
testthat::expect_snapshot_file  
testthat::expect_success  
testthat::expect_output  
testthat::expect_gte  
testthat::expect_gt  
testthat::expect_known_output  
testthat::expect_vector  
testthat::expect_identical  
testthat::expect_no_match
```

```
>biocthis::*  
  
biocthis::use_bioc_description  
biocthis::bioc_style  
biocthis::use_bioc_vignette  
  
biocthis::use_bioc_citation  
biocthis::use_bioc_github_action  
biocthis::use_bioc_pkg_templates  
biocthis::biocthis_example_pkg  
biocthis::use_bioc_support  
  
biocthis::use_bioc_issue_template  
biocthis::use_bioc_readme_rmd  
biocthis::use_bioc_coc  
biocthis::use_bioc_news_md
```



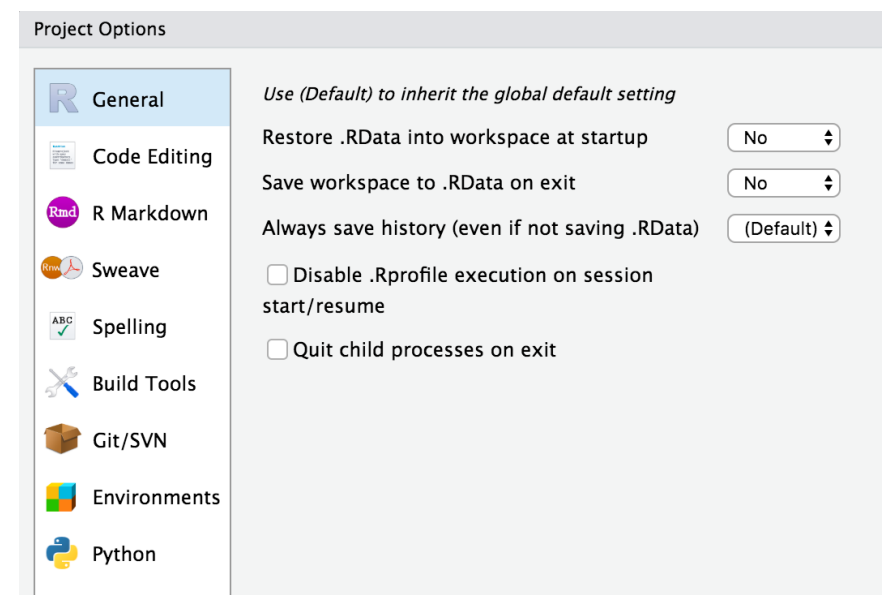
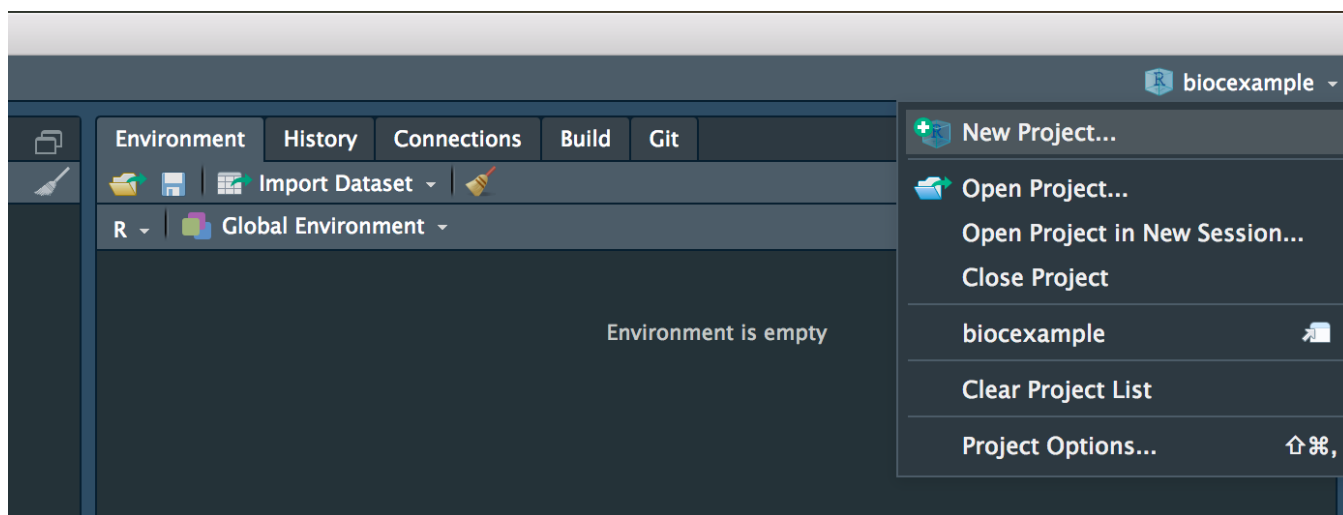
# Rstudio and package development

“devtools works hand-in-hand with RStudio, which we [Hadley] believe is the best development environment for most R users. To be clear, you can use devtools without using RStudio and you can develop packages in RStudio without using devtools. But there is a special, two-way relationship that makes it very rewarding to use devtools and RStudio together.”



# Rstudio and projects (.Rproj files)

- Projects are very “launch-able”. It’s easy to fire up a fresh instance of RStudio in a Project, with the file browser and working directory set exactly the way you need, ready for work.
- Each Project is isolated. You can have several RStudio Projects open at once and code executed in Project A does not have any effect on the R session and workspace of Project B.
- You get handy code navigation tools like F2 to jump to a function definition and Ctrl + . to look up functions or files by name.
- You get useful keyboard shortcuts and a clickable interface for common package development tasks, like generating documentation, running tests, or checking the entire package.



# Rstudio and projects (.Rproj files)

Working with projects ensures:

1. That you are not accidentally loading additional variables when opening Rstudio after rebooting
2. That your package root is always correct!
3. Many more convenient stuff you don't actually want to realize it would need to be taken care of without projects (e.g. package dependencies, versions, ...)

```
> library(devtools)
> usethis::create_package()
> usethis::use_rstudio()
> usethis::proj_sitrep()
```